ENGINEERING HANDOFF · INTERNAL

Time-on-Matter Dashboard

What we built, how it works, and the gaps we're asking engineering to close.

Date	Prepared by	Subject	Status
20 April 2026	Dillon Roberts	Alaro (RUM app `alaro-dashboard`)	Option A prototype, live

1. Executive summary

We built a standalone web dashboard that uses Datadog RUM event data — already being collected by the Alaro frontend — to answer a single question: **how is each lawyer spending their time, per matter, day by day**. It's the zero-engineering version of the E2C pipeline: we don't instrument anything new in Alaro yet; we infer time-on-matter from URLs, foreground visibility, and click targets already in the RUM stream.

FOCUSED HOURS (MAARTEN, 30D)	ATTRIBUTED TO MATTER	MATTERS TOUCHED	WORK BLOCKS
77.7h	52.5h	40	159
dedup'd, active foreground	68% of focused time	distinct folder IDs	≥15 min gap = new block

The prototype is enough to show trends ("Maarten spent 8.9h on Malbec this week; 2.7h on Magnetic Core I"), but **roughly 10–11 hours per week sits in an "unattributed" bucket** that we can't confidently tie to a matter with the current signal. This handoff describes what that bucket contains, why, and the smallest possible set of Alaro-side changes that would fix it.

2. What we built

The pieces

- **Data pipeline** (`scripts/pipeline.py`) — pulls RUM events via the Datadog API, runs an attribution pipeline, and writes `data.json` + `data.js`. Currently scoped to Maarten and

Tanya; adding users is a config change.

- **Single-file dashboard** (`dashboard.html`) — opens via `file://`, no server needed. Alaro-branded (bone palette, Grad + Aeonik Fono).
- **User picker** — Maarten / Tanya segmented control; Tanya's page shows a graceful empty state until she has RUM data.
- **Window picker** — 1d / 3d / 1w / 2w / 1m; all precomputed in one pipeline run (one API pull at 30d, in-memory slicing for smaller windows).
- **List view** — stats strip (focused / on-matter / triage / overhead / work blocks / attention ratio), hours-per-matter chart, daily stacked bar chart, full matter table, and an "unattributed time" card listing the chunks we couldn't tie to a matter, each with a Datadog Explorer deep-link.
- **Matter drill-down** (click any matter row → `#/matter/:id`) — per-matter detail view with activity-timeline Gantt, documents touched (hours, version count), AI-chat list with inferred first prompts, action-mix categorisation, heatmap (day × hour), co-touched matters, and per-session Explorer links.

Current coverage

USER	STATUS	EVENTS (30D)	SESSIONS	FOCUSED H
Maarten Schellingerhout	Live	17,074	139	77.7
Tanya	Empty state, starts logging today	0	0	0.0

3. How it works (the data flow)

Source of truth

Everything comes from **Datadog RUM**, which Alaro's frontend already emits to the EU region (`datadoghq.eu`). No DB access, no Alaro-side instrumentation beyond what's already in place.

RUM application: `alaro-dashboard`
Application ID: `db0d76f0-bf2e-4328-a098-d711035a664c`
Frontend: `dashboard.alaro.ai`
Backend (resource events): `alaro-core-api-prod-hn46mjypla-lz.a.run.app`
Session replay sample rate: 100%
Auth: `DD_API_KEY` + `DD_APP_KEY` (env)

Events we read

EVENT TYPE	WHAT WE USE IT FOR
view	Matter attribution (URL path / query), foreground time per view, doc + chat IDs
action	Click target text → matter-name inference, chat message content, action mix, activity-filter signal
session	Session duration, attention-ratio denominator, replay URL construction
resource, long_task, error	Not used by attribution; available if needed

Attribution pipeline

The pipeline runs in `attribute_time()` in `scripts/pipeline.py`:

1. **Build intervals.** For each view event, expand every `in_foreground_periods` entry into an absolute `(start, end)` tuple with a matter/triage/overhead bucket derived from `url_path` or `url_query.ideFolderId`.
2. **Merge overlaps (multi-tab dedup).** Sort by start, walk the list; each wall-clock second counts for exactly one bucket.
3. **Activity filter.** Drop foreground intervals longer than 5 min that have no action event within ±5 min — kills "tab left open" time.
4. **Aggregate.** Sum seconds per folder / per day / per bucket.
5. **Work-block grouping.** Group the per-matter event timestamps into blocks separated by ≥15 min gaps.
6. **Matter-name inference.** TF-IDF-like scoring on action target text, with "sidebar-click → navigation to matter" events weighted 3×, plus penalties for document-section vocabulary ("Agreement", "Schedule", etc.) and file extensions.
7. **Per-matter details.** For each matter, compute documents (by `docId`), AI chats (by `ideChatId`), action category mix, day×hour heatmap, co-touched matters, and session Explorer URLs.

Output

One JSON payload per user, keyed by window (1/3/7/14/30 days). Written to `data.json` + mirrored as `data.js` so `dashboard.html` works over `file://` without a local server.

What "focused time" means in this dashboard

Focused hours = the sum of `in_foreground_periods` across views, after (a) multi-tab dedup and (b) activity filtering. This is more conservative than `view.time_spent`, which inflates 3–8× because the SPA keeps views "alive" and backgrounded tabs keep ticking. Attention ratio = focused / session wall-clock ≈ 55–70% for Maarten, consistent with someone multitasking across apps during a real workday.

4. Known gaps

These are the reasons the numbers aren't pinpoint, and what Alaro-side changes would tighten each. Tagged by severity: CRITICAL for ones that visibly distort the headline numbers, HIGH, MEDIUM, LOW, OUTSIDE ALARO for Gmail/DocuSign-class items.

4.1 Home-page attribution hole

CRITICAL

Alaro's SPA router occasionally sets `url_path = "/"` with no `ideFolderId` in `url_query`. RUM captures whatever the address bar says, so those views land in the "overhead" bucket even though the user is clearly mid-matter. For Maarten this is roughly **9.3h / week** — about 35% of his focused time.

PROPOSED FIX

In the router (or wherever `currentMatterId` / `currentClientId` is tracked), add:

```
datadogRum.setGlobalContextProperty("matter_id", currentMatterId);
datadogRum.setGlobalContextProperty("client_id", currentClientId);
```

Clear the properties when neither is active. Every RUM event (view, action, resource) will then carry `@context.matter_id`; the dashboard stops inferring from URLs entirely. Estimated implementation: single-digit hours.

4.2 Foreground time ≠ active work time

HIGH

`in_foreground_periods` tells us the tab is visible, not that the user is engaged. Stare at a visible tab during a Zoom call, and it still counts. Today we partially mitigate by dropping foreground intervals ≥5 min with no action within ±5 min — but Alaro's SPA fires views very frequently, so individual intervals rarely trip that threshold.

PROPOSED FIX

Emit a lightweight heartbeat on an idle-aware timer (e.g. reset on `keydown` / `mousemove` / `scroll`, pause on 30s of no input):

```
datadogRum.addAction("heartbeat", { active: true });   // every 30s while active
datadogRum.addAction("heartbeat", { active: false });  // once, when going idle
```

The dashboard then only credits time that sits between `active:true` heartbeats. Eliminates "tab visible but ignored" time.

4.3 Matter names are inferred, not authoritative

MEDIUM

Our TF-IDF + sidebar-click heuristic gets most names right ("Malbec VDR", "ElevenLabs", "Magnetic Core I", "Marriott Oxford") but slips on sparse matters where the winning label

is a UI fragment ("Real Estate", "Personal") or a doc section. The authoritative `folder_id → folder_name` map lives in Alaro's database; we're not using it.

PROPOSED FIX (CHOOSE ONE)

1. **Periodic CSV export** — an admin script that dumps `(folder_id, folder_name, client_id, client_name, matter_type, created_at, closed_at)` on a cron; dashboard reads it as a lookup overlay.
2. **Internal admin endpoint** — `GET /api/v1/admin/folders` returning the same payload; dashboard pulls it alongside the RUM query.

4.4 No activity classification

MEDIUM

We can count clicks but not tell the difference between "reading a DD report" and "drafting a section" and "reviewing a client doc" purely from target text. Useful for management-level mix metrics ("Maarten spends 40% of matter time in AI chat"), not critical for minutes.

PROPOSED FIX

Tag key actions at emit time:

```
datadogRum.addAction("doc_edit",    { doc_id, activity: "drafting" });
datadogRum.addAction("chat_submit", { chat_id, activity: "ai_chat" });
datadogRum.addAction("doc_view",    { doc_id, activity: "reading" });
```

Only needed on the handful of core interactions.

4.5 Missing E2C milestones

HIGH

For a real E2C metric (first email received → case closed), only **work_in_progress** is visible today — the rest need either a custom action inside Alaro or an integration with an external system.

MILESTONE	TODAY	WHAT WE NEED
first_email_received	Invisible	OUTSIDE ALARO Gmail → Datadog forwarder
triage_complete	Inferrable from `/client/inbox` → `/folder/...`	Explicit `addAction("triage_complete")` on Alaro's triage confirmation
engagement_signed	Invisible	OUTSIDE ALARO DocuSign webhook → Datadog
work_in_progress	Visible (folder views)	Already covered

MILESTONE	TODAY	WHAT WE NEED
draft_delivered	Partially inferrable (Upload / Promote actions)	`addAction("draft_delivered", { doc_id, client_id })`
client_sign_off	Invisible	OUTSIDE ALARO DocuSign / email
case_closed	Invisible	`addAction("case_closed", { folder_id })` on the close button

4.6 Pre-auth resource events miss usr.email

LOW

26% of events — almost entirely `resource` events — have no `usr.email` because they fire before the user is identified. Views / actions / sessions are ~100% identified, so this doesn't affect the dashboard. If we ever want to attribute API cost / latency to specific users, fix this.

> **PROPOSED FIX**
>
> Move the `datadogRum.setUser({ id, email, name })` call earlier in the auth bootstrap, before the first API call fires.

4.7 Session replay URLs expire before event data

LOW

Datadog RUM keeps event data for 30+ days (we query up to 30d successfully) but session replay retention is shorter on most tiers. The dashboard now uses RUM Explorer URLs (not replay URLs) so deep-links keep working after replay has purged.

5. Engineering punch-list

Ordered roughly by "biggest win per hour of engineering". The first three are the ones that actually matter for the dashboard to trust its own numbers; the rest are nice-to-haves or external integrations.

#	ASK	GAP IT CLOSES	SEVERITY
1	Set `datadogRum.setGlobalContextProperty("matter_id", …)` and `("client_id", …)` wherever matter context changes (router / workspace switcher). Clear on home page.	4.1 Home-page hole	CRITICAL
2	Add explicit milestone actions: `addAction("triage_complete")`, `("draft_delivered")`, `("case_closed")`. Fire them from the UI moments that semantically match those milestones.	4.5 E2C milestones	HIGH

#	ASK	GAP IT CLOSES	SEVERITY
3	Expose a folder-metadata lookup: `GET /api/v1/admin/folders` returning `(folder_id, folder_name, client_id, client_name, matter_type, created_at, closed_at)`. Or equivalent CSV drop.	4.3 Matter names	MEDIUM
4	Emit an idle-aware heartbeat action (`addAction("heartbeat", { active })`) every 30s while the user is active; pause on idle.	4.2 Foreground ≠ active	HIGH
5	Tag core interactions with an `activity` attribute (drafting / reading / ai_chat / reviewing).	4.4 Activity class	MEDIUM
6	Reorder auth bootstrap so `datadogRum.setUser(…)` fires before the first API call.	4.6 Pre-auth events	LOW
7	(Product call, not engineering urgent.) Decide on Gmail + DocuSign event forwarding to Datadog for the outside-Alaro milestones.	4.5 External milestones	OUTSIDE

Items 1, 2, and 4 alone would take expected accuracy from roughly **±25–35%** on per-matter minutes to roughly **±5–10%** and would turn the 4 Alaro-internal E2C milestones into first-class timestamps we can query directly.

6. Code and how to run it

FILE	ROLE
`scripts/dd.py`	Datadog API client: pagination, rate-limit backoff, env auth
`scripts/pipeline.py`	Main pipeline: pull events, attribute time, infer names, compute per-matter details, write JSON
`dashboard.html`	Single-file UI: layout, Chart.js charts, matter detail drill-down, hash routing
`data.json` / `data.js`	Pipeline output, read by dashboard. Gitignored (contains client names)
`findings.md`	Original scouting report — full investigation notes and Options A/B/C
`samples/`	Raw Datadog event JSON dumps used for verification. Gitignored
`assets/fonts/`, `assets/logo/`	Alaro brand fonts and logo

Regenerate

```
cd ~/datadog-dashboard
.venv/bin/python scripts/pipeline.py
# Then refresh dashboard.html in a browser.
```

Add a new user

Pass `--users` (comma-separated emails):

```
.venv/bin/python scripts/pipeline.py --users "maarten.schellingerhout@alaro.ai,tanya@alaro.ai,new.person@alaro.ai"
```

Display name / role / `starts_on` live in `USER_PROFILES` at the top of `pipeline.py`.

Repository

`github.com/DJR100/datadog-dashboard`

7. Glossary

RUM	Real User Monitoring. Datadog's browser SDK capturing view/action/session/resource events.
view event	Emitted by the RUM SDK when a route changes. Carries URL, query params, `in_foreground_periods`.
foreground time	Time the tab is actually visible. Our base signal for "active" time; sum of `in_foreground_periods[].duration` across all views.
matter	A folder in Alaro. Keyed by `folder_id` UUID. Maps loosely to a client engagement.
work block	A continuous stretch of activity on one matter; splits on gaps ≥ 15 minutes.
triage	Time on `/client`, `/client/inbox`, or `/client/:id` – client-management work not yet tied to a specific matter.
overhead	Focused foreground time we couldn't attribute – typically `/` with no folder context. Target of fix 4.1.
attention ratio	Focused hours ÷ total session wall-clock. How much of "session time" was actually in foreground.
E2C	End-to-end: time from first client email to case closed.